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Course: CSA5117
Cryptography and network security

Assignment
UNIT - 1

AS/100
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1 Encrypt the message "DEFEND THE WALL" using caesar cipher with key=3.

Decrypt the message "KHOOZ ZRUOG" using caesar cipher with key=3.

A caesar cipher is used to encrypt the word "SECRET" to "VHFUHW". Determine the key used, and decrypt "WKLV LV D WHVW".

(a) Encrypt "DEFEND THE WALL" using caesar cipher (key=3)

Shift each letter 3 positions forward.

plaintext D E F E N D T H E W A L L

ciphertext G H I H Q G W K H Z D O O

ch ciphertext: GHIHQG WKH ZDOO

(b) Decrypt "KHOOZ ZRUOG" (key=3)

Shift each letter 3 positions back word

K H O O R Z R U O G

H E L L O W O R L D

plaintext Hello WORLD

(c) Determine the key and decrypt "WKLV LV

SECRET $\rightarrow$ VHFVHIN

~~If~~ we take keys

WKLV LV DWHVIN
THIS IS A TEST

plaintext: This is a test

2

Using the substitution key;

A $\rightarrow$ Q, B $\rightarrow$ W, C $\rightarrow$ E, D $\rightarrow$ R, E $\rightarrow$ T, F $\rightarrow$ Y, ..., Z $\rightarrow$ M

Encrypt the word "FIRE"

Given this cipher text "UTHHD QEB NRFZH YOLTK",
and knowing it is encrypted with a monoalphabetic
substitution, Find the plaintext using frequency analysis.

A $\rightarrow$ Q, B $\rightarrow$ W, C $\rightarrow$, D $\rightarrow$ R, E $\rightarrow$ T, F $\rightarrow$ Y, G $\rightarrow$ U, H $\rightarrow$
I $\rightarrow$ O, J $\rightarrow$ P, K $\rightarrow$ A, L $\rightarrow$ S, M $\rightarrow$ D, N $\rightarrow$ F, O $\rightarrow$ G, P $\rightarrow$
Q $\rightarrow$ J, R $\rightarrow$ K, S $\rightarrow$ L, T $\rightarrow$ Z, U $\rightarrow$ X, V $\rightarrow$ C, W $\rightarrow$ V, X $\rightarrow$
Y $\rightarrow$ N, Z $\rightarrow$ M

F $\rightarrow$ Y

I $\rightarrow$ O

R $\rightarrow$ K

E $\rightarrow$ T

chipser text: YOKT

(b) Decrypt "UTHHD QEB NRFZH YOLTK" using frequency analysis

Most likely plaintext:

THE QUICK BROWN FOX

Reason:

- Most frequent letters correspond to common English letter
- pattern matches a well-known English phrase.

plaintext: THE QUICK BROWN FOX

3 Encrypt the message "ATTACK AT PALACE" using the keyword "LEMON".

Decrypt the cipher text "LXFOPVEFRNHR" using the keyword "LEMON".

Given the encrypted message "QWXPXEPKMAEGTU" using the Vigenere cipher and keyword "KEY".

Find the original message.

Keyword = LEMON

Repeated keyword:

LEMONLEMONLEMO

plaintext

ATTACK AT PALACE

(a) Encrypt

ATTACK AT PALACE

LEMONLEMONLEMO

LX FOPVE FDNWEDS : cipher text

(b) Decrypt "LXFOPVEFRNHR"
using keyword LEMON

Plaintext: ATTACKATDAWN

(c) Decrypt "QNXEPKMAEETU" using keyword "KEY"

Repeat keyword:

KEYKEYKEYKEY

Decrypting gives:

plaintext: GDTALACIWUCTK

4 Given a 2×2 Key Matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

• Encrypt the message "HI" using Hill cipher

• If the message "PD" was encrypted to "QE" using the above Hill cipher matrix, verify the result by showing the matrix multiplication.

Key Matrix

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

(a) Encrypt "HI"

$$H = 7$$

$$I = 8$$

$$\text{Vector } P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$K P = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

Modulo 26

$$45 \bmod 26 = 19$$

$$54 \bmod 26 = 2$$

$$19 = T$$

$$2 = C$$

Chipher-text: TC

(b) Verify "PO" $\rightarrow$ "QC"

$$P = 15$$

$$O = 14$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 87 \\ 100 \end{bmatrix}$$

$$87 \bmod 26 = 9 = J$$

$$100 \bmod 26 = 22 = W$$

Result = JW

5 Encrypt the message "MEETMEATERLUNCH" using the key word "ZEBRAS".

Decrypt the message "EANMTEFCUHETREL" using the key word "BOARD".

(a) Encrypt "MEETMEATERLUNCH"

Key word = ZEBRAS

Arrange text:

Z E B R A S
M E E T M E
A F T E R L
U N C H x x

Alphabetical order of keyword:
A B E R S Z

Column order:

5 3 2 4 6 1

Read columns:

A → M R X

B → E T C

E → E F N

R → T E H

S → E L X

Z → M A U

Chiper text: M R X E T C E F N T E H E L X M A U

(b) Decrypt "E A N I M T E F C U H E T R E L" using key Word "BOARD"
keyword:

A, B, D, O, R

Reconstruct the columns and rearrange them to the original keyword order.

Reading row-wise gives

Plaintext: M E E T A F T E R L U N C H

6 Using the one-time pad key XMCKL, encrypt the message "HELLO".

Explain why the message "ATTACK NOW" encrypted using OTP with key "QWERTYUIO" is unbreakable under ideal conditions.

(a) Encrypt the message "HELLO" using OTP key "XMCKL"

HELLO

7 4 11 11 14

XMCKL

23 12 2 10 11

Encryption formula

$$C = (P + K) \bmod 26$$

P = 7 4 11 11 14

K = 23 12 2 10 11

C = 4 16 13 21 25

C - E O N V Z

ciphertext: EONVZ

(b)

1. The key is completely random

2. The key is as long as the plaintext

3. Each key is used only once

4. The key is kept secret between sender and receiver

7 Encrypt the message:

plain-text: "WE ARE DISCOVERED"

key word: "ZEBRA"

Decrypt the message

Ciphertext: "EVLNEACDTKESE AQR OFOJDEECUXWIN"

key: "TRIAL"

(a) WEAREDISCOVERED

Key word:

ZEBRA

Arrange into rows

Z E B R A

W E A R E

D I S C O

V E R E D

Alphabetical order of keyword:

A → column 5

B → column 3

E → column 2

R → column 4

Z → column 1

Read columns in this order

A → EOD

B → ASR

E → EIE

R → RCE

Z → WDU

chipher text: EODASREIE RCE WDU

(b) Decrypt

ciphertext:

EVLNEACDTKESEAQR OFDTDEECUXNN

key word:

TRIAL

steps:

1. Number the key word alphabetically
2. Divide ciphertext into equal column lengths.
3. Rearrange columns to original order.
4. Read row-wise.

plain text:

WE ARE DISCOVERED FLEE AT ONCE

8 You are given the following cipher text encrypted using a Monoalphabetic substitution cipher UZQSOVUOHXMO-PVGPQZWSZOPFPE SXUZUETUCG.

1. perform frequency analysis
2. Use standard English letter frequencies to hypothesize probable mappings.
3. Attempt to partially decrypt the message.
4. Provide the best guess for the plaintext message

UZQSOVUOHXMO-PVGPQZWSZOPFPE SXUZUETUCG

Frequency Count

Letter	Frequency
Z	High
O	High
P	High
U	High
S	Medium
V	Medium
E	Medium

Most frequent letters are likely to represent:

E

T

A

O

Probable Mapping

Z $\rightarrow$ E

O $\rightarrow$ T

P $\rightarrow$ A

U $\rightarrow$ O

Partial Decryption

Using the estimated mappings, some words become partially readable, while others remain unclear because the complete substitution alphabet is unknown.

you intercepted a message encrypted with a monoalphabetic Substitution cipher:

ciphertext: BJTW QTW YMTW NX FXNRUQTW

you know the original plaintext starts with:

"THIS IS THE".

Task:

1. Use the known-plaintext portion to find substitution mappings.
2. Complete the decryption of the full message.
3. Validate if any letters repeat and adjust your Substitution table if necessary.

chiphtext: BJTW QTW YMTW NX FXNRUQTW

known plain-text:

THIS IS THE

1. find the substitution mappings

<u>cipher</u>	<u>plain</u>
B	T
J	H
W	I
Q	S
Y	E
M	T

2. Continue the decryption

THIS IS THE SIMPLEST

Validation

Final plaintext: THIS IS THE SIMPLEST

10 you are given the cipher text below, encrypted using the Vigenere cipher with an unknown short key (2-5 letter long):

Cipher text: ZICVTWQNGRZGVTWAVZHCOYGLMEJ

Try to recover the key and decrypt the full message.

Cipher text: ZICVTWQNGRZGVTWAVZHCOYGLMEJ

Estimate key length

using repeated patterns and the index of coincidence, the likely key length is 3.

Recover the key

frequency analysis of each key position suggests the key
KEY

Decrypt

using the vigenere decryption formula:

$$P = (C - K) \bmod 26$$

plaintext:

THIS IS A VIGENERE CIPHER TEXT

AM